

Natural History





About the Course

Learners encounter botanical families, endangered species, invasive species, and secondary succession as they consider how man interacts with and thinks about the natural world. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 5

About Science: Grade 5

In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

Science: Grade 5

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just “feels right.” Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



Placement & Combining Tips

About Science: Grade 5

In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

Natural History: Grade 5

For approximately fifth-grade students based on a balance of complexity and reading level. However, learners may be freely combined, or sequence reordered based on needs and preferences.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
5	Natural History: Grade 5 1 time/week 15 min	Beetle Busters Nature All Around: Plants Shanley's Quest Card Game

GRADE	SCHEDULE INFO.	BOOKS
		The Bat Scientists The Forest in the Tree

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 5				
Natural History: Grade 5		General Science: Grade 5 Nature Walks & Scouting: Grades 1-8		Labs: Grade 5 Nature Notebook: Grade 5



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 5

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 5

Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

Natural History: Grade 5

- ☐ Term 1: a botanical garden or plant conservatory
- ☐ Term 2: evening walks and/or a cave, a wildlife rehabilitator, or bat conservationist
- ☐ Term 3: a forestry museum or tour

Term Prep & Teacher Tips

Science: Grade 5

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

Natural History: Grade 5

- ☐ Term 1: one or more wildflowers from p.461-508
- ☐ Term 2: bats p.241-244
- ☐ Term 3: one or more trees from p.628-692, one or more insects from p.301-395

- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
 - flower pot with soil, or place in the ground
 - trowel
 - globe/wall map

Reminders

Natural History: Grade 5

□ Note that term 1 of Natural History begins with planting a small garden, so be sure to have those supplies and any others indicated on the supply list ready. For indoor situations, the choices are many, provided there is a sunny window. For outdoors in a cooler season, some good options are parsley, carrots, or peas (may need a trellis). Basil, thyme, or beans (may need a trellis) are good options for outdoor planting in a warmer season. Note that some plants will not reach the flowering stage for a while.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Natural History: Grade 5



Beetle Busters



Nature All Around: Plants



Shanleya's Quest Card Game



The Bat Scientists



The Forest in the Tree



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Natural History: Grade 5



Household Items - Natural History: Grade 5



Quick Links

Related Course Quick Links

- ∞ [Extra Helpings](#)
- ∞ [Outdoor Work](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for iOS](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [Foundations \(See Section 13: Science\)](#)

Click THIS text
or scan the QR
code for links.



Natural History: Grade 5

How To Teach



Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



Read

- Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.



Narrate

- Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.
- Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.
- Teachers may take turns to model.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.
- Notice if there were any dates that they want to keep for their Book of Centuries.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.
 - These can be viewed as alternative ways to engage.
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Natural History: Grade 5

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 15m Natural History: Grade 5 - Lesson 1

My Own Garden

Materials: Nature All Around: Plants, video links, seeds, pot with soil or place in the ground, trowel, water

PREP: Read Teacher Tip

→ INTRO

Did you ever think of all the world as a garden full of many different plants? What about plants stands out to you most? Look at the picture on the cover of this book. How many different plants do you see? Do you recognize any of them?

→ READ, NARRATE, & DISCUSS

p.4-6 "Do you have" - "four basic parts." Take a moment to diagram a plant with these 4 parts.

p.8 "Flowering plants start" - "They shrivel up."

→ VIEW

Let's watch the beginning of the plant life cycle, and then we'll begin our own gardens, choose one or both based on time:

∞ Video Link: Growing Basil Time Lapse (1:42)

∞ Video Link: Wildflower Garden Time Lapse (3:31)

→ ACTIVITY

Begin your own small garden in the ground or even in a single pot. Plant seeds according to the seed packet and plan to check on them and water them regularly. The soil should be moist, but not wet. If it seems wet, then don't add more water. If you gently blow across the surface and the soil has a dusty character, then give it some water. As seeds sprout over the next few weeks, make careful observations in your nature journal.

→ NARRATE & DISCUSS

Use a nature journal to record what you planted.

★ TEACHER TIP

The livingness of this book lies in its artwork, so read the assigned pages, but know that some students may learn more from reading only a part and instead exploring and spending time with the pictures. When students narrate, they are analyzing the "big idea."

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2 15m Natural History: Grade 5 - Lesson 2

We Depend on Plants

Materials: Nature All Around: Plants, video links

→ INTRO

What do you recall from our last reading about plants? Why do you think plants are important to us? Plants make the air we breathe; they shape our communities and traditions; they even make many of our most important medicines. Let's find out how they do the first of these in our book.

→ READ, NARRATE, & DISCUSS

p.9 "A plant's green leaves" - "even at night."

●Add to last week's diagram or sketch a new one, showing how the plant releases oxygen into the air.

→ VIEW

∞ Video Link: Ethnobotanist (to 3:03)

∞ Video Link: Life-saving Medicines (to 4:13)

→ NARRATE & DISCUSS

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 1

WEEK 3 15m Natural History: Grade 5 - Lesson 3

Getting to Know Flowering Plants

Materials: Nature All Around: Plants, video link, Shanleya's Quest card game

→ INTRO

Recall what we learned in the last lesson. What do we call someone who studies plants? What do they actually do?

→ VIEW

∞ Video Link: What is a Botanist? (6:31)

→ READ, NARRATE, & DISCUSS

p.10-11 "Take a look" - "from the flowers."

- If you have the Shanleya's Quest card game, then explore the deck today. Notice differences in the 8 plant families presented and how the key characteristics are repeated in different plants in the family. If you do not have the card game, then refer to the link (about halfway down the webpage).

∞ Website Link: Shanleya's Quest Patterns in Plants

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4 15m Natural History: Grade 5 - Lesson 4

Caring for My Garden

Materials: The Forest in the Tree, plantings from earlier in term

→ OBSERVE

Do your seedlings appear similar to or different from each other? Are some growing slower or faster than others? Are the seeds that you planted monocots or dicots? How do the different parts of the plant, like the roots, leaves, and flowers, serve its purpose? (Refer to Nature All Around: Plants, if you need help.)

→ NARRATE & DISCUSS

Use a nature journal to record what you observed. Do you think the soil that plants grow in is very important, or can they grow in anything? What do you think is in soil?

→ OBSERVE & DISCUSS

Look at the picture on p.34. Read all or parts of p.35 to accompany the picture.

- Compare the relationship between plants and soil microbes to community members sharing resources.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 15m Natural History: Grade 5 - Lesson 5

Plants in Community

Materials: The Forest in the Tree, Shanleya's Quest card game

PREP: Read Teacher Tip

→ INTRO

Recall what we learned about how plants work and how they are part of a community.

→ READ, NARRATE, & DISCUSS

p.2-15 "Water wakes me" - "'Glomus,' I agree."

- How are the plant and the fungi working together?

★ TEACHER TIP

Are your learner(s) demonstrating more familiarity/interest in how plants work? More interest/ease in describing/identifying different types of plants? If not, consider practicing together more often, exploring extra helpings, or taking a field trip to a botanical garden.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 1

- Look at the picture on p.15. What happened when the fungi connected? (Notice the trees are also connected.)
- Play "memory" or "slap flower" based on student preference, if you have the Shanleya's Quest deck.

WEEK 6 15m Natural History: Grade 5 - Lesson 6

The Life of a Plant

Materials: The Forest in the Tree, Shanleya's Quest card game

→ INTRO

Recall what happened in the story last time. How do you think this community will live, grow, and change? Will they have new neighbors, new babies, illnesses, etc, just like our community? Do you think plant communities experience these parts of life? Why or why not?

→ READ, NARRATE, & DISCUSS

p.16-32 "Broma is older" - "shape the Earth."

- If you have the Shanleya's Quest deck, play the same game as last week or try "Crazy Flower" or "Wildflower Rummy."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7 15m Natural History: Grade 5 - Lesson 7

Caring for My Garden Activity

Materials: The Forest in the Tree, plantings from earlier in term, map/globe

→ OBSERVE

Spend a few minutes carefully observing your various seedlings. How has their appearance developed? Are you noticing distinctive characteristics yet?

→ NARRATE

Use a nature journal to record what you observed.

→ RECAP

What was the Forest in the Tree about? What kind of tree was Broma? (cacao) Where was Broma? Find the Amazon on the map/globe.

→ READ, NARRATE, & DISCUSS

p.37 "Theobroma cacao is" - "Ivory Coast to Ghana."

- Find Ivory Coast and Ghana. The cacao tree belongs to the mallow family, which means that the hibiscus flower and the linden tree are its cousins in places like Canada, the U.S., Finland, Korea, and New Zealand! Find these locations as time permits.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8 15m Natural History: Grade 5 - Lesson 8

Plants in Spring and Summer

Materials: Nature All Around: Plants, Shanleya's Quest card game

→ INTRO

Recall what we learned about how plants work. You may look back at the pictures to help. Have you noticed that different plants grow in different seasons?

→ READ, NARRATE, & DISCUSS

p.12-15 "One of the" - "they are absorbed."

- What plants have you collected so far? Which families do they belong

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 1

to? Do they grow and bloom primarily in the spring or summer? Where do they grow?

- If you have the Shanleya's Quest deck, play your favorite game. Is it getting easier to recognize their characteristics?

WEEK 9 15m Natural History: Grade 5 - Lesson 9

Plants in Fall and Winter

Materials: Nature All Around: Plants, Shanleya's Quest card game

→ INTRO

Tell me what you remember about plants in the spring and summer. What happens to plants in the fall and winter? Let's read and learn something new.

→ READ, NARRATE, & DISCUSS

p.16-19 "By fall, many" - "replanted each year"

- What 3 observations do you think are most helpful to identify a plant?
- What plants have you collected so far? Do you know what happens to them in the fall and winter? How do they come back the next year? How do they propagate?
- If you have the Shanleya's Quest deck, play your favorite game or try a new one.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 15m Natural History: Grade 5 - Lesson 10

Garden Niches

Materials: Nature All Around: Plants, Shanleya's Quest card game

→ INTRO

What do you recall about plants in different seasons? What about different places? Do you think the plants are different in different parts of our country?

→ READ, NARRATE, & DISCUSS

p.20-23 "If you were" - "for next year."

- Find your zone on the map on p.20-21. Have you visited any other zones? Were the plants different in those other zones? Have you visited different habitats in your own or another zone? Describe some of the plants you remember. Were there characteristics or patterns that you remember? If you haven't noticed plants in another zone, you might enjoy seeing those at the link:
∞ Video Link: Wildflowers in Death Valley (3:57)
- Play a game with Shanleya's Quest, as time permits.

→ NOTE

Explore "More Strange Plants" on p.26-27 for interest, if desired.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 15m Natural History: Grade 5 - Lesson 11

Caring for My Garden Activity

Materials: plantings from earlier in term

PREP: Read Teacher Tip

→ INTRO

Your seedlings are probably much bigger now! Spend today carefully observing them. Do your observations reveal to which families they belong yet? If not, then keep observing and be patient!

★ TEACHER TIP

When students take exams next week, evaluate what they recall, but more importantly, whether any of the new plants they have met are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less

Natural History: Grade 5

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Term 1

→ OBSERVE

→ NARRATE & DISCUSS

Use a nature journal to record what you observed.

business, etc?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12 15m Natural History: Grade 5 - Lesson 12

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:

The Forest in the Tree, Nature All Around: Plants

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 2

WEEK 1 15m Natural History: Grade 5 - Lesson 13

Working in the Dark

Materials: The Bat Scientists

PREP: Read Teacher Tip

→ INTRO

Look at the picture on the cover. Have you watched bats flying at night before? What do you know about bats? Briefly look at p.8-10 based on interest before beginning.

→ READ & NARRATE

p.13-15 "Do you see" - "his favorite."

→ READ, NARRATE, & DISCUSS

p.15-16 "Merlin Tuttle" - "to save them."

→ SUPPLEMENTAL

Estimating the number of animals in a population, like bats in a cave, is challenging! Check out the linked photos and see if you can estimate the number of animals in just a glance.

∞ Image Link: Herd of Bison

∞ Image Link: School of Fish

∞ Image Link: Murmuration

★ TEACHER TIP

Sometimes more complex readings can be broken into smaller parts. In this passage, the first part introduces us to the bats and the second to the scientist who studies them.

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2 15m Natural History: Grade 5 - Lesson 14

Watching Out for Bats

Materials: The Bat Scientists

→ INTRO

What did we read in the last passage? Where are we? What is it like? Who are we with? Today, we'll read about bats in their ecosystem.

→ READ, NARRATE, & DISCUSS

p.19-24 "The baking hot" - "fear of bats." (including p.21)

- Use Google or a local resource to discover what the bats in your area eat and what eats them. Where do they live? Who else is dependent on them?
- Discuss the food chain/web based on the reading OR for the bats in your area. Draw or diagram, if helpful.

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 15m Natural History: Grade 5 - Lesson 15

Knowing Bats

Materials: The Bat Scientists

→ INTRO

Recall what we learned in the last passage. Let's get to know the bats a bit better in this passage.

→ READ, NARRATE, & DISCUSS

p.27-33 "Want to see" - "truth about bats."

- Discuss the wing diagram on p.33. What stands out to you? Copy the diagram, if desired.

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 5

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Term 2

WEEK 4 15m Natural History: Grade 5 - Lesson 16

What's So Special About Bats?

Materials: The Bat Scientists

→ INTRO

Recall what we learned in the last passage. Can you think of any characteristics that make bats really special?

→ READ, NARRATE, & DISCUSS

p.34-39 "Bats are unique" - "second chance."

- Find the picture of a bat from your region or just a bat that you like at the link. Can you notice the adaptations discussed in today's reading? Does this tell you anything about this bat?
- ∞ Image Link: Bats by Region

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 15m Natural History: Grade 5 - Lesson 17

What Do You Do with a Problem?

Materials: video link

→ INTRO

Recall what we learned in the last passage. Today, we have a video that struggles with a bat problem, so let's watch and see what we think.

→ VIEW & NARRATE

∞ Video Link: The Case for Vampire Bats (8:48)

- What do you think? What should be done for the people? What should be done for the bats?
- What makes a creature valuable?

→ NOTE

Explore "Batty Myths" on p.38 for interest, if desired.

★ TEACHER NOTE

The required video in this lesson includes a cartoon of a vampire bat biting a child at 0:18 and mentions evolution at 2:51. Teachers might allow these to pass by the way, but should preview if they have concerns. The video could be started at 0:27 if they think the cartoon will be frightening, and teachers could decide in advance how they want to address any questions about evolution.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 15m Natural History: Grade 5 - Lesson 18

Bat Habitat

Materials: The Bat Scientists

PREP: Read Teacher Tip

→ INTRO

Let's remember what our video was about and where we left off in our book. A lot of bats live in caves. Have you ever been to a cave? Why do you think bats like caves?

→ READ, NARRATE, & DISCUSS

p.41-45 "You'd never know" - "hibernating bats like."
p.45 "How a cave traps cold air."

- Describe the cave as a habitat. Copy the diagram, if desired.

★ TEACHER TIP

Are your learner(s) demonstrating more familiarity/interest in bats or their role in the ecosystem? More interest/ease in describing/identifying different types of plants? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or conservationist that has bats.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7 15m Natural History: Grade 5 - Lesson 19

Stewardship

Materials: The Bat Scientists

PREP: Read Teacher Tip

★ TEACHER TIP

Keystone species and umbrella species are sometimes used interchangeably, but an umbrella species usually migrates, has high habitat needs, and affects a great number of species also connected

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Term 2

→ INTRO

Tell me what you learned about bats in the last reading. Knowing how important caves are, what do you think we can do to take care of them?

→ READ, NARRATE, & DISCUSS

p.45-51 "People make" - "survive, too."

- The author uses the term "umbrella species." If you are familiar with the term "keystone species," what do you think is similar or different about the idea of "umbrella species"? The habitat is an important part of this idea - discuss the difference between using something, such as a plant, animal, or cave, without regard for its goodness versus responsible stewardship.

→ NOTE

Explore "Gating Laurel Cave" on p.48-49 for interest, if desired.

to its habitat over its entire range.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8 15m Natural History: Grade 5 - Lesson 20

More Bat Habitats

Materials: The Bat Scientists

PREP: Read Teacher Tip

→ INTRO

Tell me what you learned about bats in the last reading. There are a few other kinds of homes that bats like, and some of them might surprise you.

→ READ, NARRATE, & DISCUSS

p.53-58 "A woman gently" - "busy bridge."

→ NOTE

Explore "Celebrity Bridge Bats" on p.60-61 for interest, if desired.
∞ Video Link: Austin's Nocturnal Neighbors (1:27)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9 15m Natural History: Grade 5 - Lesson 21

Stewarding Problems

Materials: The Bat Scientists

→ INTRO

Tell me what you know about where bats live. Let's find out how learning more about their habitats can help us make better choices and do the right thing.

→ READ, NARRATE, & DISCUSS

p.62-64 "Though many" - "of last refuge."

- Make up a poem about bats together!

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 15m Natural History: Grade 5 - Lesson 22

White Nose Syndrome

Materials: The Bat Scientists

→ INTRO

Tell me what you learned about bat habitat in the last reading. Bats have another challenge in addition to homelessness. It is called white nose syndrome, so let's read about that today.

→ READ, NARRATE, & DISCUSS

p.67-70 "After sleeping" - "our lifetimes."

★ TEACHER NOTE

Evolution is mentioned in the third sentence on p.68: "Everything from... is uncertain." Teachers might let it pass by the way, decide how they would like to discuss, or consider adjusting the text if reading aloud.

• OUTDOOR WORK

Weekly walks and daily outdoor

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Term 2

→ NOTE

Explore "Cleanable Caves" on p.71 for interest, if desired.

time are vital! Resources in Quick Links.

WEEK 11 15m Natural History: Grade 5 - Lesson 23

Listening to Bats

Materials: The Bat Scientists

PREP: Read Teacher Tip

→ INTRO

Remind me what we read in the last passage. Let's finish the book by learning how the bat scientists observe the bats.

→ READ, NARRATE, & DISCUSS

p.72-73 "Back in" - "it came from."

p.75-77 "Having done" - "without bats."

→ SUPPLEMENTAL

∞ Video Link: Using a Bat Detector (3:15)

∞ Video Link: Filming Bats (4:22)

→ NOTE

Explore "Killer Wind" on p.74 for interest, if desired.

★ TEACHER TIP When students take exams next week, evaluate what they recall, but more importantly, whether bats or their role in the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12 15m Natural History: Grade 5 - Lesson 24

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:
The Bat Scientists

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 3

WEEK 1 15m Natural History: Grade 5 - Lesson 25

What Happened to the Forest

Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Look at the picture on the cover. What do you notice about this insect? In this book, we will learn about the delicate ecological balance between insect and plant life and what happens when it goes awry.

→ READ & NARRATE

p.4-6 "Ryan Zumpano" - "right thing to do?"

→ READ, NARRATE, & DISCUSS

p.7-8 "People who" - "hardwood forest."

- Tell about a favorite tree.

★ TEACHER TIP

Some students will benefit from reading and narrating this selection in two parts, as shown: the first introduces us to the forest, and the second to the problem.

★ TEACHER NOTE

Evolution is mentioned in the second paragraph on p.8. Teachers might let it pass by the way, decide how they would like to discuss, or consider adjusting the text if reading aloud.

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2 15m Natural History: Grade 5 - Lesson 26

The Beetle

Materials: Beetle Busters

→ INTRO

Recall what has happened in the story. Today, we will learn more about the beetle that is causing the trouble.

→ READ, NARRATE, & DISCUSS

p.9-13 "The Asian longhorned" - "entire neighborhood."

- If cutting trees in one community would save the trees in your neighborhood, is it worth it? What if you lived in the community whose trees were being cut? What if it were your favorite tree?

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 15m Natural History: Grade 5 - Lesson 27

A Life in Rings

Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Tell me what you remember about the beetle. Can they tell if the tree has beetles inside it? What do you think it would look like on the inside? Today we are going to read a bit about how these tree scientists look inside the trees using something called dendrochronology.

→ READ, NARRATE, & DISCUSS

p.13-14 "Lots of things" - "the tree itself."

→ SUPPLEMENTAL

∞ Website Link: Dendrochronology at Mesa Verde

★ TEACHER TIP

Dendrochronology comes from the Greek words meaning "tree" and "time."

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 3

WEEK 4 15m Natural History: Grade 5 - Lesson 28

How the Beetle Got Here

Materials: Beetle Busters, globe

→ INTRO

Recall what dendrochronology is. Where do you think this beetle came from, and how did it get to North America? Look at your globe and find China. Then trace with your finger a path to North America.

→ READ, NARRATE, & DISCUSS

p.15-20 "The story of" - "in the country."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 15m Natural History: Grade 5 - Lesson 29

An Infestation

Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Tell me the story so far of how the beetle got to North America. What does it matter if this beetle is in North America?

→ READ, NARRATE, & DISCUSS

Ch.2 p.21-24 "In August 2008" - "we have to lose."

- Knowing what a healthy tree should look like helped the scientist notice a sick tree quickly. Look at the picture in the Tree ID section on p.24. Do you know the leaf, fruit, twig, and bark of a tree in your place? Describe what you know.

★ TEACHER TIP

Are your learners demonstrating more familiarity/interest in native/invasive species or their role in the ecosystem? More interest/ease in describing/identifying particular native/invasive species? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 15m Natural History: Grade 5 - Lesson 30

Working Together

Materials: Beetle Busters

→ INTRO

Remind me what we read about in the last passage. This sounds like a really big problem, so let's find out what they are going to do about it.

→ READ, NARRATE, & DISCUSS

p.25-30 "Clint McFarland" - "easier to bear."

- Explain how it helped them to first understand the beetle's life cycle.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7 15m Natural History: Grade 5 - Lesson 31

Beetle Busting

Materials: Beetle Busters

→ INTRO

Recall how they decided to problem-solve the beetle infestation. Let's read more about the process today.

→ READ, NARRATE, & DISCUSS

p.31-34 "Joining a tree" - "thirty years?"

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 3

WEEK 8 15m Natural History: Grade 5 - Lesson 32

Learning from the Forest

Materials: Beetle Busters

→ INTRO

Tell me what you remember from the last passage. The scientists didn't want to just solve the problem, though. They wanted to learn from it, and that's what we're going to read about today.

→ READ, NARRATE, & DISCUSS

p.35-40 "Early in" - "and beyond."

→ NOTE

Explore "More Beetle Busting" on p.41-42 for interest, if desired.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9 15m Natural History: Grade 5 - Lesson 33

Analyzing Data

Materials: Beetle Busters

→ INTRO

Recall what the scientists did to learn more about the beetle infestation. They found out three things from their special study, so let's find out what they learned.

→ READ, NARRATE, & DISCUSS

p.43-46 "Back in the" - "those forests."

- How do you think the knowledge they gained helps?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 15m Natural History: Grade 5 - Lesson 34

Succession Takes a Long Time

Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Remind me what the scientists found out from their work. Let's read about how they expect the forest to recover in a process called succession.

→ READ, NARRATE, & DISCUSS

p.47-52 "Change happens" - "urban forest."

- What does the word "succession" mean? How does it make sense to describe what's happening in this situation?
- If students are familiar with succession from Life on Surtsey, help them compare/contrast the primary succession on Surtsey and the secondary succession here.

★ TEACHER TIP

Notice that this reading is about the concept of succession and what that means in regard to the infested forest. Students may narrate the sequence of events of succession or of the author's thought, but they may instead give points to describe the concept and its relevance to the story.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 15m Natural History: Grade 5 - Lesson 35

Dialogue is Key

Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Tell me how scientists expect the infested forest to recover. How do you think the people who live there feel about having all those trees cut?

★ TEACHER TIP

This is another passage in which students might access the concluding ideas better by giving points to describe the author's view rather than narrating as a sequence of events.

★ TEACHER TIP When students take exams next week, evaluate what they recall, but more

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Term 3

→ READ, NARRATE, & DISCUSS

p.55-58 "But to the" - "I believe in them."

- Explain both points of view about whether or not the trees should be cut.
- What do you think about the problem of whether or not to cut down the infested trees? How can they make a good decision?

importantly, whether a plant/animal or their role in the ecosystem is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12 15m Natural History: Grade 5 - Lesson 36

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:
Beetle Busters

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Examination

- Term 1** · The Forest in the Tree & Nature All Around: Plants: Tell about or draw what you know about how plants work. OR Compare in word or picture three different plant types that you found this term.
- Term 2** · The Bat Scientists: Explain in words or pictures what bats need to thrive in an ecosystem. OR Tell about or draw three unique characteristics of the bat.
- Term 3** · Beetle Busters: Explain in words or pictures why an invasive species is bad for an ecosystem. OR Tell about or diagram one way that scientists learn about the health of a forest.